PassFrame tuition

Factors, Multiples & Primes

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: List all the factors of 12.
Method & answer: 1, 2, 3, 4, 6, 12.

Standard: Is 37 a prime number?
Method & answer: Yes — 37 has no factors other than 1 and 37, so it is prime.

Advanced: Find the highest common factor (HCF) of 18 and 24.
Method & answer: Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Common factors: 1, 2, 3, 6. HCF = 6.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. List all the factors of 8.
2. List all the factors of 15.
3. Select ALL of these that are multiples of 4: A) 12 B) 14 C) 16 D) 18 E) 20
4. List the first 5 multiples of 6.
5. Is 21 a multiple of 7?
6. Is 9 a factor of 42?
7. Is 17 a prime number? A) Yes B) No, it is divisible by 3 C) No, it is a square number D) No, it is even
8. Is 20 a prime number?
9. List all the factors of 20.
10. What is the smallest multiple of 9 that is greater than 40? A) 36 B) 45 C) 42 D) 49
11. List the prime numbers between 10 and 20.
12. Is 45 a multiple of 5?
13. Which of these is NOT a factor of 18? A) 2 B) 4 C) 6 D) 9
14. What is the first common multiple of 3 and 5 (other than 0)?
15. Is 1 a prime number? A) Yes, because it can only be divided by 1 B) No, because it has only one factor C) Yes, because it is odd D) No, because it is even

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find all the common factors of 12 and 18.
2. Find the highest common factor (HCF) of 16 and 24. A) 4 B) 6 C) 8 D) 12
3. Find the lowest common multiple (LCM) of 4 and 6.
6. Is 51 a prime number? A) Yes B) No — 51 = 3 × 17 C) No — 51 is even D) No — 51 is a square number
7. Find the first three common multiples of 2 and 5.
8. What is the highest common factor of 20 and 30?
11. Find all the factors of 48.
12. Select ALL of these that are prime: A) 23 B) 27 C) 29 D) 33
13. Find the HCF of 12, 18 and 24.

4. Find the lowest common multiple (LCM) of 3 and 8.

5. List all the factor pairs of 36.

9. Select ALL of these that are square numbers: A) 36 B) 45 C) 49 D) 60 E) 64

10. Is 64 a square number?

14. A number is a multiple of both 4 and 6. What is the smallest such number? A) 8 B) 10 C) 12 D) 24

15. Find two prime numbers that add to 30.

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the smallest number less than 50 that is a multiple of both 6 and 8.

2. I am a 2-digit prime number between 50 and 60. My digits add to 8. What number am I? A) 51 B) 53 C) 57 D) 59

3. Find the number between 60 and 66 that has exactly 2 factors (i.e. is prime).

4. The HCF of two numbers is 6 and their LCM is 36. One of the numbers is 12. What is the other number?

5. A number is both a factor of 60 and a multiple of 10. Select ALL of these that are possible values: A) 10 B) 15 C) 20 D) 30 E) 40

6. Which number between 30 and 36 is a prime number?

7. Find the smallest number that is a multiple of 3, 4 and 5.

8. Two numbers have HCF 4 and both are less than 30. Which of these could be the pair? A) 4 and 4 B) 8 and 12 C) 6 and 9 D) 10 and 15

9. A square number is also a factor of 100. Which numbers work?

10. I am thinking of a number. It is a multiple of 7, a factor of 84, greater than 20, less than 30, and even. What number am I?

11. Which prime numbers multiply together to make 30? A) 2, 3 and 5 B) 2 and 15 C) 3 and 10 D) 5 and 6

12. Find a number less than 30 that has exactly 4 factors.

13. The LCM of two numbers is 24. One number is 8. What could the other number be?

14. Select ALL of the factors of 45 that are also prime: A) 3 B) 5 C) 9 D) 15 E) 45

15. A rectangle has an area of 48 cm² and whole-number sides. What are the possible pairs of side lengths?